

Statistics & Probability

ANSWER KEY



Measures of center

- 1 Find the mean of 12, 15, 18, 9, 21.

$$\text{Sum} = 75. \text{ Mean} = \frac{75}{5} = 15.$$

- 2 Find the median of 4, 9, 2, 11, 7, 15.

$$\text{Sorted: } 2, 4, 7, 9, 11, 15. \text{ Median} = \frac{7 + 9}{2} = 8.$$

- 3 Find the mode of 3, 7, 5, 7, 2, 7, 9.

The value 7 appears most often, so the mode is 7.

Measures of spread

- 4 Find the range of the data set 22, 18, 35, 9, 27.

$$\text{Range} = 35 - 9 = 26.$$

- 5 Find the interquartile range (IQR) of 4, 7, 9, 12, 15, 18, 21.

Lower half (excluding the median 12): 4, 7, 9, so $Q_1 = 7$. Upper half: 15, 18, 21, so $Q_3 = 18$. $\text{IQR} = 18 - 7 = 11$.

- 6 A data set has a range of 40 and a minimum value of 15. Find the maximum value.

$$\text{Maximum} = 15 + 40 = 55.$$

Basic probability

- 7 A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.

$$\text{Total marbles} = 10. P(\text{green}) = \frac{2}{10} = \frac{1}{5}.$$

- 8 A fair six-sided die is rolled once. Find $P(\text{even number})$.

$$\text{Even outcomes: } \{2, 4, 6\}. P = \frac{3}{6} = \frac{1}{2}.$$

- 9 A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than 5})$.

$$\text{Numbers greater than 5: } \{6, 7, 8\}. P = \frac{3}{8}.$$

Combined probability: AND / OR

- 10 A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.

$$4 \text{ kings} + 4 \text{ queens} = 8 \text{ favorable cards. } P = \frac{8}{52} = \frac{2}{13}.$$

- 11 Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$$P(A \cap B) = 0.4 \times 0.5 = 0.20.$$

- 12 A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.

$$P = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = \frac{9}{25}.$$

Probability without replacement

- 13 A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

$$P = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33}.$$

- 14 A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.

$$P = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}.$$

Data displays

- 15 The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.

Experimental $P(4) = \frac{9}{40} = 0.225$. Theoretical $P(4) = \frac{1}{6} \approx 0.167$. The experimental probability is somewhat higher than the theoretical probability, which is expected random variation in a finite sample.

Expected value

- 16 A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.

Expected winnings $= 0.1 \times 10 = 1$ dollar. Expected net profit $= 1 - 2 = -1$ dollar (an expected loss of 1 dollar per play).

- 17 A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.

$$E = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4} = 12.5 \text{ points.}$$

Effects of transformations on statistics

- 18** A data set has mean 20. If every value in the set is increased by 5, find the new mean.

$$\text{New mean} = 20 + 5 = 25.$$

- 19** A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.

$$\text{New standard deviation} = 3 \times 4 = 12.$$

A well-designed word problem: a survey of students

- 20** This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

(a) By inclusion-exclusion, the number doing at least one activity is $120 + 80 - 30 = 170$. Neither $= 200 - 170 = 30$. (b) $P(\text{sport or instrument}) = \frac{170}{200} = 0.85$. (c) $P(\text{instrument} \mid \text{sport}) = \frac{P(\text{both})}{P(\text{sport})} = \frac{30}{120} = \frac{1}{4}$.

Multiple choice — easy

- 21** Find the mean of 6, 50, 46, 36, 7.

a. 29 ✓

b. 31

c. 145

d. 27

$$\text{Sum} = 145. \text{ Mean} = \frac{145}{5} = 29.$$

22 Find the mean of 41, 43, 15, 15, 6.

a. 24 ✓

b. 22

c. 26

d. 120

$$\text{Sum} = 120. \text{ Mean} = \frac{120}{5} = 24.$$

23 Find the mean of 21, 18, 14, 35, 2.

a. 90

b. 16

c. 18 ✓

d. 20

$$\text{Sum} = 90. \text{ Mean} = \frac{90}{5} = 18.$$

24 Find the mean of 48, 28, 35, 30, 19.

a. 34

b. 160

c. 30

d. 32 ✓

$$\text{Sum} = 160. \text{ Mean} = \frac{160}{5} = 32.$$

25 Find the mean of 23, 12, 44, 5, 26.

a. 24

b. 20

c. 110

d. 22 ✓

$$\text{Sum} = 110. \text{ Mean} = \frac{110}{5} = 22.$$

26 Find the mean of 26, 25, 26, 25, 48.

a. 28

b. 32

c. 30 ✓

d. 150

$$\text{Sum} = 150. \text{ Mean} = \frac{150}{5} = 30.$$

27 Find the mean of 4, 31, 3, 26, 1.

a. 13 ✓

b. 11

c. 65

d. 15

$$\text{Sum} = 65. \text{ Mean} = \frac{65}{5} = 13.$$

28 Find the mean of 22, 6, 29, 26, 42.

a. 125

b. 25 ✓

c. 23

d. 27

$$\text{Sum} = 125. \text{ Mean} = \frac{125}{5} = 25.$$

29 Find the mean of 45, 35, 33, 19, 13.

- a. 27
- b. 145
- c. 31
- d. 29 ✓

$$\text{Sum} = 145. \text{ Mean} = \frac{145}{5} = 29.$$

30 Find the mean of 9, 21, 19, 19, 22.

- a. 18 ✓
- b. 16
- c. 20
- d. 90

$$\text{Sum} = 90. \text{ Mean} = \frac{90}{5} = 18.$$

31 Find the mean of 6, 21, 21, 28, 29.

- a. 19
- b. 105
- c. 21 ✓
- d. 23

$$\text{Sum} = 105. \text{ Mean} = \frac{105}{5} = 21.$$

32 Find the mean of 50, 23, 27, 33, 12.

- a. 145
- b. 31
- c. 29 ✓
- d. 27

$$\text{Sum} = 145. \text{ Mean} = \frac{145}{5} = 29.$$

33 Find the mean of 3, 38, 13, 31, 5.

a. 20

b. 18 ✓

c. 16

d. 90

Sum = 90. Mean = $\frac{90}{5} = 18$.

34 Find the mean of 14, 29, 33, 44, 30.

a. 28

b. 32

c. 30 ✓

d. 150

Sum = 150. Mean = $\frac{150}{5} = 30$.

35 Find the mean of 31, 26, 30, 21, 47.

a. 31 ✓

b. 29

c. 155

d. 33

Sum = 155. Mean = $\frac{155}{5} = 31$.

Multiple choice — medium

- 36 A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.

- a. $\frac{6}{23}$ ✓
- b. $\frac{8}{23}$
- c. $\frac{11}{23}$
- d. $\frac{9}{23}$

Total marbles = 23. $P(\text{red}) = \frac{6}{23}$.

- 37 A jar has 3 green, 5 blue, and 12 purple marbles. Find $P(\text{blue})$.

- a. $\frac{3}{20}$
- b. $\frac{7}{20}$
- c. $\frac{3}{5}$
- d. $\frac{1}{4}$ ✓

Total marbles = 20. $P(\text{blue}) = \frac{5}{20} = \frac{1}{4}$.

- 38 A jar has 6 red, 2 orange, and 2 blue marbles. Find $P(\text{red})$.

- a. $\frac{1}{5}$
- b. $\frac{3}{5}$ ✓
- c. $\frac{4}{5}$
- d. $\frac{9}{10}$

Total marbles = 10. $P(\text{red}) = \frac{6}{10} = \frac{3}{5}$.

- 39 A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

- a. $\frac{7}{16}$
- b. $\frac{3}{8}$
- c. $\frac{1}{4}$ ✓
- d. $\frac{1}{2}$

Total marbles = 16. $P(\text{red}) = \frac{4}{16} = \frac{1}{4}$.

40 A jar has 2 green, 1 orange, and 10 yellow marbles. Find $P(\text{orange})$.

a. $\frac{1}{13}$ ✓

b. $\frac{2}{13}$

c. $\frac{3}{13}$

d. $\frac{10}{13}$

Total marbles = 13. $P(\text{orange}) = \frac{1}{13}$.

41 A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$

b. $\frac{10}{21}$

c. $\frac{11}{21}$

d. $\frac{9}{21}$ ✓

Total marbles = 21. $P(\text{red}) = \frac{9}{21} = \frac{3}{7}$.

42 A jar has 1 yellow, 5 purple, and 5 orange marbles. Find $P(\text{yellow})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{3}{11}$

d. $\frac{1}{11}$ ✓

Total marbles = 11. $P(\text{yellow}) = \frac{1}{11}$.

43 A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

- a. $\frac{1}{8}$
- b. $\frac{1}{2}$
- c. $\frac{5}{8}$
- d. $\frac{1}{4}$ ✓

$$\text{Total marbles} = 8. P(\text{red}) = \frac{2}{8} = \frac{1}{4}.$$

44 A jar has 7 purple, 11 orange, and 10 green marbles. Find $P(\text{green})$.

- a. $\frac{11}{28}$
- b. $\frac{3}{7}$
- c. $\frac{1}{4}$
- d. $\frac{5}{14}$ ✓

$$\text{Total marbles} = 28. P(\text{green}) = \frac{10}{28} = \frac{5}{14}.$$

45 A jar has 7 yellow, 1 green, and 9 red marbles. Find $P(\text{yellow})$.

- a. $\frac{7}{17}$ ✓
- b. $\frac{9}{17}$
- c. $\frac{1}{17}$
- d. $\frac{10}{17}$

$$\text{Total marbles} = 17. P(\text{yellow}) = \frac{7}{17}.$$

46 A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

- a. $\frac{1}{3}$
- b. $\frac{8}{27}$
- c. $\frac{10}{27}$ ✓
- d. $\frac{4}{9}$

$$\text{Total marbles} = 27. P(\text{red}) = \frac{10}{27}.$$

47 A jar has 5 yellow, 9 orange, and 3 blue marbles. Find $P(\text{orange})$.

a. $\frac{9}{17}$ ✓

b. $\frac{5}{17}$

c. $\frac{11}{17}$

d. $\frac{3}{17}$

Total marbles = 17. $P(\text{orange}) = \frac{9}{17}$.

48 A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{5}{26}$

b. $\frac{1}{13}$ ✓

c. $\frac{2}{13}$

d. $\frac{6}{13}$

Total marbles = 26. $P(\text{blue}) = \frac{2}{26} = \frac{1}{13}$.

49 A jar has 7 red, 1 yellow, and 9 green marbles. Find $P(\text{red})$.

a. $\frac{10}{17}$

b. $\frac{7}{17}$ ✓

c. $\frac{9}{17}$

d. $\frac{1}{17}$

Total marbles = 17. $P(\text{red}) = \frac{7}{17}$.

50 A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

- a. $\frac{5}{11}$
- b. $\frac{9}{22}$ ✓
- c. $\frac{3}{22}$
- d. $\frac{1}{2}$

Total marbles = 22. $P(\text{blue}) = \frac{9}{22}$.

Multiple choice — hard

51 Two independent events have $P(A) = 0.1$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

- a. 0.28
- b. 0.13
- c. 0.18
- d. 0.08 ✓

$P(A \cap B) = 0.1 \times 0.8 = 0.08$.

52 Two independent events have $P(A) = 0.1$ and $P(B) = 0.7$. Find $P(A \text{ and } B)$.

- a. 0.27
- b. 0.17
- c. 0.12
- d. 0.07 ✓

$P(A \cap B) = 0.1 \times 0.7 = 0.07$.

53 Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. 0.32 ✓

b. 0.42

c. 0.52

d. 0.22

$$P(A \cap B) = 0.8 \times 0.4 = 0.32.$$

54 Two independent events have $P(A) = 0.5$ and $P(B) = 0.6$. Find $P(A \text{ and } B)$.

a. 0.40

b. 0.20

c. 0.50

d. 0.30 ✓

$$P(A \cap B) = 0.5 \times 0.6 = 0.30.$$

55 Two independent events have $P(A) = 0.8$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 0.40 ✓

b. 0.30

c. 0.50

d. 0.60

$$P(A \cap B) = 0.8 \times 0.5 = 0.40.$$

56 A bag has 3 red and 10 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{3}{13}$

b. $\frac{15}{26}$

c. $\frac{1}{26}$ ✓

d. $\frac{5}{13}$

$$P = \frac{3}{13} \times \frac{2}{12} = \frac{6}{156} = \frac{1}{26}.$$

- 57 A bag has 7 red and 9 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

- a. $\frac{7}{16}$
- b. $\frac{21}{40}$
- c. $\frac{3}{10}$
- d. $\frac{7}{40}$ ✓

$$P = \frac{7}{16} \times \frac{6}{15} = \frac{42}{240} = \frac{7}{40}.$$

- 58 A bag has 6 red and 6 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

- a. $\frac{5}{22}$ ✓
- b. $\frac{1}{2}$
- c. $\frac{6}{11}$
- d. $\frac{31}{132}$

$$P = \frac{6}{12} \times \frac{5}{11} = \frac{30}{132} = \frac{5}{22}.$$

- 59 A bag has 7 red and 4 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

- a. $\frac{21}{55}$ ✓
- b. $\frac{28}{55}$
- c. $\frac{7}{11}$
- d. $\frac{6}{55}$

$$P = \frac{7}{11} \times \frac{6}{10} = \frac{42}{110} = \frac{21}{55}.$$

- 60 A bag has 5 red and 3 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{3}{28}$

b. $\frac{5}{14}$ ✓

c. $\frac{15}{28}$

d. $\frac{5}{8}$

$$P = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}.$$

- 61 A spinner has 4 equal sections labeled 15, 20, 35, and 45 points. Find the expected value of one spin.

a. 28.75 ✓

b. 29.75

c. 27.75

d. 31.75

$$E = \frac{15 + 20 + 35 + 45}{4} = \frac{115}{4} = 28.75 \text{ points.}$$

- 62 A spinner has 4 equal sections labeled 10, 20, 25, and 40 points. Find the expected value of one spin.

a. 26.75

b. 24.75

c. 23.75 ✓

d. 22.75

$$E = \frac{10 + 20 + 25 + 40}{4} = \frac{95}{4} = 23.75 \text{ points.}$$

- 63 A spinner has 4 equal sections labeled 5, 25, 30, and 35 points. Find the expected value of one spin.

a. 26.75

b. 22.75

c. 24.75

d. 23.75 ✓

$$E = \frac{5 + 25 + 30 + 35}{4} = \frac{95}{4} = 23.75 \text{ points.}$$

64 A spinner has 4 equal sections labeled 15, 20, 30, and 40 points. Find the expected value of one spin.

a. 27.25

b. 26.25 ✓

c. 29.25

d. 25.25

$$E = \frac{15 + 20 + 30 + 40}{4} = \frac{105}{4} = 26.25 \text{ points.}$$

65 A spinner has 4 equal sections labeled 5, 15, 35, and 45 points. Find the expected value of one spin.

a. 28

b. 24

c. 26

d. 25 ✓

$$E = \frac{5 + 15 + 35 + 45}{4} = \frac{100}{4} = 25 \text{ points.}$$